Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Contents

Integrals

1.1 Medium Integral

Symmetric Exponential Rational Integral

Evaluate the definite integral:

$$\int_{-1}^{1} e^{x^2}\left[\frac{x^2}{2-x}+\frac{1}{(2+x)^2}\right]dx$$

Solution: Let the given integral be denoted as I:

$$I=\int_{-1}^{1} e^{x^2}\left[\frac{x^2}{2-x}+\frac{1}{(2+x)^2}\right]dx$$

We split the integral into two separate terms:

$$I=\int_{-1}^{1} e^{x^2}\frac{x^2}{2-x}\,dx+\int_{-1}^{1} e^{x^2}\frac{1}{(2+x)^2}\,dx$$

Notice that the interval of integration $[-1,1]$ is symmetric about the origin. In the first integral, we perform the reflection substitution $x\to -x$ (since $\int_{-1}^{1} g(x)\,dx=\int_{-1}^{1} g(-x)\,dx$):

$$\int_{-1}^{1} e^{x^2}\frac{x^2}{2-x}\,dx=\int_{-1}^{1} e^{(-x)^2}\frac{(-x)^2}{2-(-x)}\,dx=\int_{-1}^{1} e^{x^2}\frac{x^2}{2+x}\,dx$$

Substituting this reflected form back into the expression for I, both terms now share the common denominator base $(2+x)$:

$$I=\int_{-1}^{1} e^{x^2}\left[\frac{x^2}{2+x}+\frac{1}{(2+x)^2}\right]dx$$

We now evaluate the second term using integration by parts. Let:

$$u=e^{x^2}\implies du=2xe^{x^2}\,dx$$

$$dv=\frac{1}{(2+x)^2}\,dx\implies v=-\frac{1}{2+x}$$

Applying the integration by parts formula $\int u\,dv=uv-\int v\,du$:

$$\int_{-1}^{1} e^{x^2}\frac{1}{(2+x)^2}\,dx=\left[-\frac{e^{x^2}}{2+x}\right]_{-1}^{1}-\int_{-1}^{1}\left(-\frac{1}{2+x}\right)\left(2xe^{x^2}\right)dx$$

$$=\left[-\frac{e^{x^2}}{2+x}\right]_{-1}^{1}+\int_{-1}^{1} e^{x^2}\frac{2x}{2+x}\,dx$$

Evaluating the boundary term:

$$\left[-\frac{e^{x^2}}{2+x}\right]_{-1}^{1} = \left(-\frac{e^{1^2}}{2+1}\right) - \left(-\frac{e^{(-1)^2}}{2+(-1)}\right) = -\frac{e}{3} + e = \frac{2e}{3}$$

Substituting this integration by parts expansion back into the total integral I:

$$I = \int_{-1}^{1} e^{x^2} \frac{x^2}{2+x}\, dx + \frac{2e}{3} + \int_{-1}^{1} e^{x^2} \frac{2x}{2+x}\, dx$$

We combine the two remaining integrals under a single integrand:

$$I = \frac{2e}{3} + \int_{-1}^{1} e^{x^2} \left[\frac{x^2+2x}{2+x}\right] dx$$

Factoring the numerator of the rational term reveals an exact algebraic cancellation:

$$\frac{x^2+2x}{2+x} = \frac{x(x+2)}{2+x} = x$$

Thus, the integral simplifies dramatically to:

$$I = \frac{2e}{3} + \int_{-1}^{1} x e^{x^2}\, dx$$

Observe that the integrand $h(x) = xe^{x^2}$ is an odd function since $h(-x) = -xe^{(-x)^2} = -h(x)$. The definite integral of an odd function over any symmetric interval $[-a, a]$ is identically zero:

$$\int_{-1}^{1} x e^{x^2}\, dx = \left[\frac{1}{2} e^{x^2}\right]_{-1}^{1} = \frac{1}{2}e - \frac{1}{2}e = 0$$

Therefore, all integral terms vanish completely, yielding the exact closed-form value:

$$I = \frac{2e}{3}$$

1.2 Hard Integral

Reciprocal Exponent Hyperbolic Integral

Evaluate the definite integral:

$$\int_{-\infty}^{\infty} \frac{e^{x+\frac{1}{x}}}{\left(e^x + e^{\frac{1}{x}}\right)^2}\, dx$$

Solution: Let the integral over the entire real line be denoted by I:

$$I = \int_{-\infty}^{\infty} \frac{e^{x+\frac{1}{x}}}{\left(e^x + e^{\frac{1}{x}}\right)^2}\, dx$$

We begin by simplifying the algebraic structure of the integrand. Expanding the squared binomial in the denominator:

$$\left(e^x + e^{\frac{1}{x}}\right)^2 = e^{2x} + 2e^{x+\frac{1}{x}} + e^{\frac{2}{x}}$$

Dividing both the numerator and the denominator by the term $e^{x+\frac{1}{x}}$, we obtain:

$$\frac{e^{x+\frac{1}{x}}}{e^{2x} + 2e^{x+\frac{1}{x}} + e^{\frac{2}{x}}} = \frac{1}{e^{x-\frac{1}{x}} + 2 + e^{-\left(x-\frac{1}{x}\right)}}$$

Notice that the denominator is a perfect square of exponentials in terms of the variable $u(x) = x - \frac{1}{x}$:

$$e^u + 2 + e^{-u} = \left(e^{u/2} + e^{-u/2}\right)^2 = \left(2\cosh\left(\frac{u}{2}\right)\right)^2 = 4\cosh^2\left(\frac{u}{2}\right)$$

Therefore, our integral can be written concisely in terms of the hyperbolic secant function:

$$I = \frac{1}{4}\int_{-\infty}^{\infty} \operatorname{sech}^2\left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right) dx$$

To evaluate this integral without an explicit $1 + \frac{1}{x^2}$ differential factor, we exploit Cauchy-Schlömilch (or Glasser's) transformation properties. Let us split the integral into positive and negative half-lines:

$$I = \frac{1}{4}\int_{-\infty}^{0} \operatorname{sech}^2\left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right) dx + \frac{1}{4}\int_{0}^{\infty} \operatorname{sech}^2\left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right) dx$$

For the integral over the negative half-line, let $x = -y$ where $y \in (0, \infty)$. Since $\operatorname{sech}^2(-z) = \operatorname{sech}^2(z)$ is an even function:

$$\int_{-\infty}^{0} \operatorname{sech}^2\left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right) dx = \int_{0}^{\infty} \operatorname{sech}^2\left(\frac{1}{2}\left(-y + \frac{1}{y}\right)\right) dy = \int_{0}^{\infty} \operatorname{sech}^2\left(\frac{1}{2}\left(y - \frac{1}{y}\right)\right) dy$$

Thus, the integral is symmetric over both halves of the real domain:

$$I = \frac{1}{2}\int_{0}^{\infty} \operatorname{sech}^2\left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right) dx$$

We now apply the inversion substitution $x = \frac{1}{t} \implies dx = -\frac{1}{t^2}\,dt$ to this positive half-line integral:

$$\int_{0}^{\infty} \operatorname{sech}^2\left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right) dx = \int_{\infty}^{0} \operatorname{sech}^2\left(\frac{1}{2}\left(\frac{1}{t} - t\right)\right)\left(-\frac{1}{t^2}\right) dt = \int_{0}^{\infty} \frac{1}{x^2}\operatorname{sech}^2\left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right) dx$$

Averaging the original expression and its inverted form yields:

$$\int_{0}^{\infty} \operatorname{sech}^2\left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right) dx = \frac{1}{2}\int_{0}^{\infty} \left(1 + \frac{1}{x^2}\right) \operatorname{sech}^2\left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right) dx$$

Substituting this equivalence back into our equation for I:

$$I = \frac{1}{4}\int_{0}^{\infty} \left(1 + \frac{1}{x^2}\right) \operatorname{sech}^2\left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right) dx$$

Now we execute the direct substitution $u = x - \frac{1}{x}$, which gives the differential $du = \left(1 + \frac{1}{x^2}\right) dx$. As $x \to 0^+$, $u \to -\infty$, and as $x \to \infty$, $u \to \infty$:

$$I = \frac{1}{4} \int_{-\infty}^{\infty} \operatorname{sech}^2 \left(\frac{u}{2}\right) du$$

Finally, let $w = \frac{u}{2} \implies du = 2\,dw$:

$$I = \frac{1}{4} \int_{-\infty}^{\infty} \operatorname{sech}^2(w)(2\,dw) = \frac{1}{2} \left[\tanh(w)\right]_{-\infty}^{\infty} = \frac{1}{2} \left(1 - (-1)\right) = \frac{1}{2}(2) = 1$$

The exact value of the improper integral is:

$$I = 1$$

Differentials

2.1 Medium Differential

Non-Linear Homogeneous First-Order Equation

Compute $f(e)$ if:

$$xf'(x) = f(x) + \sqrt{x^2 - f^2(x)}$$

Given the initial condition $f(1) = 0$.

Solution: We begin by dividing both sides of the differential equation by the independent variable x (for $x > 0$):

$$f'(x) = \frac{f(x)}{x} + \sqrt{1 - \left(\frac{f(x)}{x}\right)^2}$$

We recognize this as a homogeneous first-order differential equation. We introduce the rational substitution $v(x) = \frac{f(x)}{x}$, which implies $f(x) = xv(x)$. Differentiating with respect to x via the product rule yields:

$$f'(x) = v + x\frac{dv}{dx}$$

Substituting this expression into our differential equation:

$$v + x\frac{dv}{dx} = v + \sqrt{1 - v^2}$$

Subtracting v from both sides simplifies the equation to:

$$x\frac{dv}{dx} = \sqrt{1 - v^2}$$

We separate variables to isolate v and x:

$$\frac{dv}{\sqrt{1 - v^2}} = \frac{dx}{x}$$

Integrating both sides of the separated equation:

$$\int \frac{dv}{\sqrt{1 - v^2}} = \int \frac{dx}{x} \implies \arcsin(v) = \ln|x| + C$$

Substituting back $v = \frac{f(x)}{x}$:

$$\arcsin\left(\frac{f(x)}{x}\right) = \ln|x| + C$$

We now apply the initial condition $f(1) = 0$ to determine the constant of integration C:

$$\arcsin\left(\frac{0}{1}\right) = \ln(1) + C \implies 0 = 0 + C \implies C = 0$$

Thus, the specific relation governing the function is:

$$\arcsin\left(\frac{f(x)}{x}\right) = \ln|x|$$

Taking the sine of both sides (valid for domains where $|\ln|x|| \leq \frac{\pi}{2}$):

$$\frac{f(x)}{x} = \sin(\ln|x|) \implies f(x) = x\sin(\ln|x|)$$

To find the required value $f(e)$, we evaluate this function at $x = e$:

$$f(e) = e\sin(\ln(e)) = e\sin(1)$$

Since $\ln(e) = 1$ radian $\approx 57.3^\circ < \frac{\pi}{2}$, this point lies well within the principal branch of the inverse sine function. The exact value is:

$$f(e) = e\sin(1)$$

2.2 Hard Differential

Dirac Delta Impulse Second-Order Equation

Solve the second-order linear differential equation:

$$y'' + 4y' + 5y = \delta(t - 2\pi)$$

Given the initial conditions $y(0) = 0$ and $y'(0) = 0$, find the exact value of $y\left(\frac{5\pi}{2}\right)$.

Solution: We solve this impulse-driven initial value problem using the Laplace transform. Let $Y(s) = \mathcal{L}\{y(t)\}$ denote the Laplace transform of $y(t)$. Taking the Laplace transform of both sides of the differential equation:

$$\mathcal{L}\{y''\} + 4\mathcal{L}\{y'\} + 5\mathcal{L}\{y\} = \mathcal{L}\{\delta(t - 2\pi)\}$$

Using the operational formulas for derivatives and the Dirac delta function:

$$\left[s^2Y(s) - sy(0) - y'(0)\right] + 4\left[sY(s) - y(0)\right] + 5Y(s) = e^{-2\pi s}$$

Substituting the zero initial conditions $y(0) = 0$ and $y'(0) = 0$:

$$\left(s^2 + 4s + 5\right)Y(s) = e^{-2\pi s}$$

We solve algebraically for $Y(s)$ in the s-domain and complete the square in the denominator:

$$Y(s) = \frac{e^{-2\pi s}}{s^2 + 4s + 5} = \frac{e^{-2\pi s}}{(s+2)^2 + 1}$$

We recognize that without the exponential delay factor, the inverse Laplace transform of the rational function is a damped sine wave:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2+1}\right\} = e^{-2t}\sin(t)$$

By the Second Shifting Theorem (time-delay property), multiplying by $e^{-2\pi s}$ shifts the time-domain function by $t_0 = 2\pi$ and activates it via the Heaviside step function $u(t-2\pi)$:

$$y(t) = \mathcal{L}^{-1}\left\{e^{-2\pi s}F(s)\right\} = e^{-2(t-2\pi)}\sin(t-2\pi)u(t-2\pi)$$

By the 2π-periodicity of the sine function, we have $\sin(t-2\pi) = \sin(t)$. Therefore, for all time $t \geq 2\pi$:

$$y(t) = e^{-2(t-2\pi)}\sin(t)$$

We are required to evaluate the solution at the time instant $t = \frac{5\pi}{2}$. Since $\frac{5\pi}{2} = 2.5\pi > 2\pi$, the step function is active ($u = 1$):

$$y\left(\frac{5\pi}{2}\right) = e^{-2\left(\frac{5\pi}{2}-2\pi\right)}\sin\left(\frac{5\pi}{2}\right)$$

Computing the exponent and the trigonometric term:

$$\frac{5\pi}{2} - 2\pi = \frac{\pi}{2} \implies -2\left(\frac{\pi}{2}\right) = -\pi$$

$$\sin\left(\frac{5\pi}{2}\right) = \sin\left(2\pi + \frac{\pi}{2}\right) = \sin\left(\frac{\pi}{2}\right) = 1$$

Substituting these values yields the final exact result:

$$y\left(\frac{5\pi}{2}\right) = e^{-\pi}(1) = e^{-\pi}$$

Derivatives

3.1 Medium Derivative

Inverse Function Derivative of Integral Function

Compute $(f^{-1}(0))'$ if:

$$f(x) = \int_{1/x}^{x} \frac{\sin(xt)}{xt}\,dt$$

Solution: By the standard derivative theorem for inverse functions, the derivative of $f^{-1}(y)$ evaluated at $y = 0$ is related to the reciprocal of the derivative of $f(x)$:

$$(f^{-1})'(0) = \frac{1}{f'(x_0)} \qquad \text{where } f(x_0) = 0$$

Step 1: Determine the root x_0 where $f(x_0) = 0$ We examine the limits of the definite integral defining $f(x)$. A definite integral evaluates to zero when its lower and upper limits of integration are identical:

$$\frac{1}{x_0} = x_0 \implies x_0^2 = 1 \implies x_0 = 1 \quad (\text{assuming domain } x > 0)$$

Checking directly at $x = 1$:

$$f(1) = \int_{1/1}^{1} \frac{\sin(t)}{t}\,dt = \int_{1}^{1} \frac{\sin(t)}{t}\,dt = 0$$

Thus, $x_0 = 1$, and we need to compute $(f^{-1})'(0) = \frac{1}{f'(1)}$.

Step 2: Compute the first derivative $f'(x)$ We can evaluate $f'(x)$ using two independent methods to ensure complete analytical rigor.

Method A: Leibniz Integral Rule The Leibniz rule for differentiating an integral with variable limits and integrand is:

$$\frac{d}{dx}\int_{a(x)}^{b(x)} g(x,t)\,dt = g(x,b(x))b'(x) - g(x,a(x))a'(x) + \int_{a(x)}^{b(x)} \frac{\partial g}{\partial x}(x,t)\,dt$$

For our function $f(x)$, we have $g(x,t) = \frac{\sin(xt)}{xt}$, $a(x) = \frac{1}{x}$, and $b(x) = x$. When evaluating $f'(x)$ specifically at $x = 1$, notice that since $a(1) = b(1) = 1$, the integral term vanishes completely:

$$\int_{a(1)}^{b(1)} \frac{\partial g}{\partial x}(1,t)\,dt = \int_{1}^{1} \frac{\partial g}{\partial x}(1,t)\,dt = 0$$

We only need to evaluate the boundary terms at $x = 1$:

$$f'(1) = \left[\frac{\sin(xt)}{xt}\right]_{t=x=1} \cdot \frac{d}{dx}(x)\Big|_{x=1} - \left[\frac{\sin(xt)}{xt}\right]_{t=1/x=1} \cdot \frac{d}{dx}\left(\frac{1}{x}\right)\Big|_{x=1}$$

$$f'(1) = \left(\frac{\sin(1)}{1}\right)(1) - \left(\frac{\sin(1)}{1}\right)(-1) = \sin(1) + \sin(1) = 2\sin(1)$$

Method B: Variable Substitution We substitute $u = xt \implies dt = \frac{du}{x}$. The limits change to $u = x\left(\frac{1}{x}\right) = 1$ and $u = x(x) = x^2$:

$$f(x) = \int_1^{x^2} \frac{\sin(u)}{u}\left(\frac{du}{x}\right) = \frac{1}{x}\int_1^{x^2} \frac{\sin(u)}{u}\, du$$

Differentiating via the product rule and Fundamental Theorem of Calculus:

$$f'(x) = -\frac{1}{x^2}\int_1^{x^2} \frac{\sin(u)}{u}\, du + \frac{1}{x}\left(\frac{\sin(x^2)}{x^2} \cdot 2x\right) = -\frac{1}{x^2}\int_1^{x^2} \frac{\sin(u)}{u}\, du + \frac{2\sin(x^2)}{x^2}$$

Evaluating at $x = 1$:

$$f'(1) = -1\int_1^{1} \frac{\sin(u)}{u}\, du + \frac{2\sin(1)}{1} = 0 + 2\sin(1) = 2\sin(1)$$

Step 3: Evaluate the inverse derivative Substituting $f'(1) = 2\sin(1)$ into our formula yields:

$$(f^{-1})'(0) = \frac{1}{2\sin(1)} = \frac{1}{2}\csc(1)$$

3.2 Hard Derivative

Alternating Hyperbolic Arcsine Series Derivative

Given the function:

$$f(x) = \sum_{n=1}^{\infty} (-x)^n \left(\frac{2^n (n!)^2}{(2n+1)!}\right)$$

Find $f''(1)$.

Solution: Let us define the generating coefficient $c_n = \frac{2^n(n!)^2}{(2n+1)!}$ and examine the auxiliary power series $S(z) = \sum_{n=0}^{\infty} c_n z^n$. Notice that our function $f(x)$ is simply related to $S(z)$ by setting $z = -x$ and subtracting the $n = 0$ term ($c_0 = 1$):

$$f(x) = S(-x) - 1 \implies f'(x) = -S'(-x) \implies f''(x) = S''(-x)$$

To evaluate $f''(1) = S''(-1)$, we derive an exact differential equation satisfied by $g(x) = S(-x) = f(x) + 1$. We establish the recurrence relation between consecutive coefficients c_n and c_{n+1}:

$$c_{n+1} = \frac{2^{n+1}((n+1)!)^2}{(2n+3)!} = c_n \cdot \frac{2(n+1)^2}{(2n+3)(2n+2)} = c_n \cdot \frac{2(n+1)^2}{2(n+1)(2n+3)} = c_n \cdot \frac{n+1}{2n+3}$$

This gives the linear coefficient relation:

$$(2n+3)c_{n+1} = (n+1)c_n$$

In our alternating series $g(x) = \sum_{n=0}^{\infty} c_n(-1)^n x^n = \sum_{n=0}^{\infty} a_n x^n$, let $a_n = c_n(-1)^n$. Then $a_{n+1} = -c_{n+1}(-1)^n$, which transforms the recurrence into:

$$(2n+3)a_{n+1} + (n+1)a_n = 0$$

Multiplying this relation by x^{n+1} and summing over all integers $n \geq 0$:

$$\sum_{n=0}^{\infty}(2n+3)a_{n+1}x^{n+1} + \sum_{n=0}^{\infty}(n+1)a_n x^{n+1} = 0$$

We express both summations in terms of the function $g(x)$ and its derivative $g'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1}$:

- First summation (shifting index to $m = n+1$):

$$\sum_{m=1}^{\infty}(2m+1)a_m x^m = 2x\sum_{m=1}^{\infty} m a_m x^{m-1} + \sum_{m=1}^{\infty} a_m x^m = 2xg'(x) + g(x) - a_0$$

  Since $a_0 = c_0 = 1$, this equals $2xg'(x) + g(x) - 1$.

- Second summation:

$$x\sum_{n=0}^{\infty} n a_n x^n + x\sum_{n=0}^{\infty} a_n x^n = x^2 g'(x) + xg(x)$$

Adding the two summations together yields the first-order differential equation:

$$\left(x^2 + 2x\right) g'(x) + (x+1)g(x) = 1$$

To find the second derivative $g''(1)$, we differentiate this entire equation with respect to x:

$$(2x+2)g'(x) + \left(x^2+2x\right) g''(x) + g(x) + (x+1)g'(x) = 0$$

$$\left(x^2+2x\right) g''(x) + (3x+3)g'(x) + g(x) = 0$$

Evaluating this second-order relation at $x = 1$:

$$\left(1^2 + 2(1)\right) g''(1) + (3(1)+3)g'(1) + g(1) = 0 \implies 3g''(1) + 6g'(1) + g(1) = 0$$

$$g''(1) = -2g'(1) - \frac{1}{3}g(1)$$

We now determine the exact numerical values of $g(1)$ and $g'(1)$. The closed-form summation of our alternating series is known in terms of the inverse hyperbolic sine function:

$$g(x) = S(-x) = \frac{2\operatorname{arsinh}\left(\sqrt{\frac{x}{2}}\right)}{\sqrt{x(x+2)}}$$

Evaluating at $x = 1$:

$$g(1) = \frac{2\operatorname{arsinh}\left(\frac{1}{\sqrt{2}}\right)}{\sqrt{3}}$$

Using the logarithmic definition $\text{arsinh}(w) = \ln\left(w + \sqrt{w^2+1}\right)$:

$$\text{arsinh}\left(\frac{1}{\sqrt{2}}\right) = \ln\left(\frac{1}{\sqrt{2}} + \sqrt{\frac{3}{2}}\right) = \ln\left(\frac{1+\sqrt{3}}{\sqrt{2}}\right) = \frac{1}{2}\ln\left(2+\sqrt{3}\right)$$

Therefore, the exact value of $g(1)$ is:

$$g(1) = \frac{\ln\left(2+\sqrt{3}\right)}{\sqrt{3}}$$

Next, we evaluate our first-order differential equation at $x = 1$ to find $g'(1)$:

$$\left(1^2 + 2(1)\right) g'(1) + (1+1)g(1) = 1 \implies 3g'(1) + 2g(1) = 1 \implies g'(1) = \frac{1-2g(1)}{3}$$

Substituting this $g'(1)$ into our equation for $g''(1)$:

$$g''(1) = -2\left(\frac{1-2g(1)}{3}\right) - \frac{1}{3}g(1) = -\frac{2}{3} + \frac{4}{3}g(1) - \frac{1}{3}g(1) = g(1) - \frac{2}{3}$$

Since $f''(x) = g''(x)$, we obtain our final closed-form exact solution:

$$f''(1) = \frac{\ln\left(2+\sqrt{3}\right)}{\sqrt{3}} - \frac{2}{3} \approx 0.093679$$

Physics & Engineering

4.1 Mechanics

Pendulum Clock Thermal Time Expansion Error

A pendulum clock keeps correct time at a temperature 15°C and has a time period of 2 s. The coefficient of linear expansion of the rod is $\lambda = 1.25 \times 10^{-5}\ \mathrm{K}^{-1}$. If the temperature rises to 25°C, what is the clock's time error after one day, in seconds?

Solution: The time period T of a simple pendulum of length L in a gravitational field g is given by:

$$T = 2\pi\sqrt{\frac{L}{g}}$$

When the temperature increases by $\Delta\theta = \theta_2 - \theta_1 = 25°\mathrm{C} - 15°\mathrm{C} = 10\ \mathrm{K}$, the metallic rod undergoes linear thermal expansion. Its new length L' becomes:

$$L' = L(1 + \lambda\Delta\theta)$$

Consequently, the new oscillation time period T' expands to:

$$T' = 2\pi\sqrt{\frac{L(1+\lambda\Delta\theta)}{g}} = T(1+\lambda\Delta\theta)^{1/2}$$

Since the thermal expansion term is extremely small ($\lambda\Delta\theta \ll 1$), we apply the first-order binomial approximation $(1+x)^{1/2} \approx 1 + \frac{1}{2}x$:

$$T' \approx T\left(1 + \frac{1}{2}\lambda\Delta\theta\right)$$

The fractional change in the period of the clock is therefore directly proportional to temperature rise:

$$\frac{\Delta T}{T} = \frac{T' - T}{T} \approx \frac{1}{2}\lambda\Delta\theta$$

Because the period increases ($T' > T$), each oscillation cycle takes longer to complete. Therefore, the clock ticks slower than true time and **loses time**. Over a total duration $t = 1$ day $= 24$ hours $\times$ 3600 s/hour $= 86,400$ seconds, the accumulated time error Δt is:

$$\Delta t = t \cdot \frac{\Delta T}{T} = t \cdot \left(\frac{1}{2}\lambda\Delta\theta\right)$$

Substituting the numerical values into the formula:

$$\Delta t = 86,400\ \mathrm{s} \times \frac{1}{2} \times \left(1.25 \times 10^{-5}\ \mathrm{K}^{-1}\right) \times 10\ \mathrm{K}$$

$$\Delta t = 86,400 \times 0.5 \times 1.25 \times 10^{-4} = 43,200 \times 1.25 \times 10^{-4}$$

$$\Delta t = 54,000 \times 10^{-4} = 5.4\ \text{seconds}$$

After one full day at 25°C, the clock loses exactly:

$$\Delta t = 5.4\ \text{seconds}$$

4.2 Quantum Mechanics

Relativistic Dimuon Invariant Mass Kinematics

Two muons are detected in a proton-proton collision. Their measured four-momenta are given as follows:

- Muon 1: $E_1 = 53.4\,\text{GeV}$, $p_{x1} = 18.2\,\text{GeV}/c$, $p_{y1} = 41.5\,\text{GeV}/c$, $p_{z1} = 28.9\,\text{GeV}/c$.
- Muon 2: $E_2 = 42.0\,\text{GeV}$, $p_{x2} = -14.7\,\text{GeV}/c$, $p_{y2} = -39.3\,\text{GeV}/c$, $p_{z2} = -25.4\,\text{GeV}/c$.

Calculate the invariant mass M of the dimuon system in GeV/c^2. Assume the muon mass is negligible. Use $M^2 = (E_1 + E_2)^2 - |\vec{p}_1 + \vec{p}_2|^2$.

Solution: We calculate the invariant mass of the combined two-particle system using relativistic four-momentum conservation:

$$M^2c^4 = E_{\text{tot}}^2 - |\vec{p}_{\text{tot}}|^2c^2$$

In natural particle physics units where $c = 1$, this simplifies to $M^2 = E_{\text{tot}}^2 - |\vec{p}_{\text{tot}}|^2$.

Step 1: Calculate Total System Energy E_{tot}

$$E_{\text{tot}} = E_1 + E_2 = 53.4\,\text{GeV} + 42.0\,\text{GeV} = 95.4\,\text{GeV}$$

Squaring the total energy:

$$E_{\text{tot}}^2 = (95.4)^2 = 9,101.16\,\text{GeV}^2$$

Step 2: Calculate Total Momentum Components $\vec{p}_{\text{tot}} = \vec{p}_1 + \vec{p}_2$ We sum the linear momentum vectors component-by-component along each Cartesian axis:

- $p_{x,\text{tot}} = p_{x1} + p_{x2} = 18.2 + (-14.7) = 3.5\,\text{GeV}/c$
- $p_{y,\text{tot}} = p_{y1} + p_{y2} = 41.5 + (-39.3) = 2.2\,\text{GeV}/c$
- $p_{z,\text{tot}} = p_{z1} + p_{z2} = 28.9 + (-25.4) = 3.5\,\text{GeV}/c$

Step 3: Calculate Total Momentum Magnitude Squared $|\vec{p}_{\text{tot}}|^2$

$$|\vec{p}_{\text{tot}}|^2 = p_{x,\text{tot}}^2 + p_{y,\text{tot}}^2 + p_{z,\text{tot}}^2 = (3.5)^2 + (2.2)^2 + (3.5)^2$$

$$|\vec{p}_{\text{tot}}|^2 = 12.25 + 4.84 + 12.25 = 29.34\,(\text{GeV}/c)^2$$

Step 4: Compute Invariant Mass Squared M^2 Subtracting the squared momentum from the squared energy:

$$M^2 = E_{\text{tot}}^2 - |\vec{p}_{\text{tot}}|^2 = 9,101.16 - 29.34 = 9,071.82\,(\text{GeV}/c^2)^2$$

Step 5: Determine Invariant Mass M Taking the principal square root yields:

$$M = \sqrt{9,071.82} \approx 95.2461\,\text{GeV}/c^2$$

Rounding to three significant figures, the invariant mass of the dimuon system is:

$$M \approx 95.2\,\text{GeV}/c^2$$

(Note: This value is characteristic of high-mass dilepton events produced in collider experiments, lying near the Z boson mass peak at $\sim 91.2\,\text{GeV}/c^2$.)

4.3 Electromagnetism

Electromagnetic Railgun Launcher Dynamics

An electromagnetic launcher consists of two parallel conducting rails and a projectile of mass $m = 3.20$ kg. The inductance $L(x)$ of the system varies with position x:

$$L(x) = 0.50 + 0.80x + 0.15x^2 \text{ mH}$$

A constant current $I = 2500$ A is maintained. The projectile starts from rest at $x = 0$ and leaves at $x = 4.0$ m. The force is $F = \frac{1}{2}I^2\frac{dL}{dx}$. Find the exit velocity in m s^{-1}.

Solution: We determine the exit velocity of the projectile by applying the Work-Energy Theorem. The total mechanical work W done by the electromagnetic propulsion force as the projectile travels from its initial position $x_0 = 0$ to its exit position $x_f = 4.0$ m is:

$$W = \int_{x_0}^{x_f} F(x)\,dx = \int_0^{4.0} \frac{1}{2}I^2\frac{dL}{dx}\,dx$$

Because the driving current $I = 2500$ A is maintained constant by the power supply throughout the launch, the factor $\frac{1}{2}I^2$ can be factored out of the integral:

$$W = \frac{1}{2}I^2\int_0^{4.0}\frac{dL}{dx}\,dx = \frac{1}{2}I^2\left[L(4.0) - L(0)\right] = \frac{1}{2}I^2\Delta L$$

We evaluate the total change in system inductance $\Delta L = L(4.0) - L(0)$:

- Initial inductance at $x = 0$:

$$L(0) = 0.50 \text{ mH}$$

- Final inductance at $x = 4.0$ m:

$$L(4.0) = 0.50 + 0.80(4.0) + 0.15(4.0)^2 = 0.50 + 3.20 + 0.15(16) = 0.50 + 3.20 + 2.40 = 6.10 \text{ mH}$$

The net inductance increase across the barrel is:

$$\Delta L = 6.10 \text{ mH} - 0.50 \text{ mH} = 5.60 \text{ mH} = 5.60 \times 10^{-3} \text{ H}$$

Now we compute the total magnetic work W delivered to the projectile:

$$W = \frac{1}{2}(2500 \text{ A})^2 \times \left(5.60 \times 10^{-3} \text{ H}\right)$$

Noting that $(2500)^2 = 6.25 \times 10^6$ A^2:

$$W = \frac{1}{2}\left(6.25 \times 10^6\right)\left(5.60 \times 10^{-3}\right) = \frac{1}{2}(6.25 \times 5.60) \times 10^3 \text{ J}$$

$$W = \frac{1}{2}(35.0) \times 10^3 \text{ J} = 17,500 \text{ Joules}$$

By the Work-Energy Theorem, since the projectile begins from rest ($v_0 = 0$), the net work equals its final kinetic energy at barrel exit:

$$W = \Delta K = \frac{1}{2}mv_f^2 - 0 \implies \frac{1}{2}mv_f^2 = 17,500 \text{ J}$$

Solving algebraically for the exit velocity v_f with projectile mass $m = 3.20$ kg:

$$mv_f^2 = 2W = 35,000 \text{ J} \implies v_f^2 = \frac{35,000}{3.20} = 10,937.5 \text{ (m s}^{-1})^2$$

Taking the square root yields the final launch velocity:

$$v_f = \sqrt{10,937.5} \approx 104.58 \text{ m s}^{-1}$$

Rounding to three significant figures, the projectile leaves the launcher at:

$$v_f \approx 105 \text{ m s}^{-1}$$